



A distance-based weighting framework for boosting the performance of dynamic ensemble selection

Zhong-Liang Zhang*, Yu-Yu Chen, Jing Li, Xing-Gang Luo

School of Management, Hangzhou Dianzi University, Hangzhou 310018, China

ARTICLE INFO

Keywords:

Multiple classifier system
Classifier competence
Dynamic weighting
Dynamic ensemble selection
Classifier fusion

ABSTRACT

Dynamic Ensemble Selection (DES) strategy is one of the most common and effective techniques in machine learning to deal with classification problems. DES systems aim to construct an ensemble consisting of the most appropriate classifiers selected from the candidate classifier pool according to the competence level of the individual classifier. Since several classifiers are selected, their combination becomes crucial. However, most of current DES approaches focus on the combination of the selected classifiers while ignoring the local information surrounding the query sample needed to be classified. In order to boost the performance of DES-based classification systems, we in this paper propose a dynamic weighting framework for the classifier fusion during obtaining the final output of an DES system. In particular, the proposed method first employs a DES approach to obtain a group of classifiers for a query sample. Then, the hypothesis vector of the selected ensemble is obtained based on the analysis of consensus. Finally, a distance-based weighting scheme is developed to adjust the hypothesis vector depending on the closeness of the query sample to each class. The proposed method is tested on 30 real-world datasets with six well-known DES approaches based on both homogeneous and heterogeneous ensemble. The obtained results, supported by proper statistical tests, show that our method outperforms, both in terms of accuracy and kappa measures, the original DES framework.

1. Introduction

Supervised classification learning is one of the most important techniques in machine learning and data mining, which receives a large amount of attention from the research committee (Woźniak, Graña, & Corchado, 2014). In a classification problem, a set of instances called training set is given, where each instance consists of several features or attributes. One of the attributes called the decision attribute should be categorical indicates the class to which each instance belongs and the others called input attributes are either continues or categorical. The aim of a classification task is to build a model of classifying the decision attribute according to the input attributes. A classification task occurs when an unknown pattern is needed to be assigned into one of the pre-defined classes based on the input attributes related to that instance. Classification techniques have extensive application in many real-world fields, including problems in fault detection (Capozzoli, Lauro, & Khan, 2015), biometrics (Goswami, Mittal, Majumdar, Vatsa, & Singh, 2016), face recognition (Juefei-Xu & Savvides, 2016), text classification (Rehman, Javed, & Babri, 2017), medical diagnosis (Peker, 2016), and sentiment classification (Onan, Korukoğlu, & Bulut, 2017).

Although a large number of classification learning algorithms have been proposed, different classifiers have unique classification power both for theoretical and experimental results reported in the literature (Dietterich, 2000; Woźniak et al., 2014). In the past

* Corresponding author.

E-mail address: zzl19860210@126.com (Z.-L. Zhang).

decade, Multiple Classifier Systems (MCSs) that combine different classifiers have been proven as an effective and robust strategy for classification problems. The relative techniques and approaches for MCSs can be roughly divided into two different categories: static classifier systems and dynamic classifier systems. In the former strategy, the system employ the same classifier for all the query samples, for example, bootstrap aggregating (Bagging) (Breiman, 1996), boosting (Freund & Schapire, 1995; 1996), and random subspaces (Ho, 1998). While in the latter scheme, the classifier system named dynamic classifier system aims to select the most appropriate classifiers for each unknown pattern, for instance, dynamic classifier selection (DCS) (Giacinto & Roli, 2000; Woods, Kegelmeyer, & Bowyer, 1997) and dynamic ensemble selection (DES) (Ko, Sabourin, & Britto, 2008; Woźniak et al., 2014).

DCS strategy attempts to find the best classifier for predicting a given sample, since different classifiers usually have different performance on different region. To do so, the key component is the estimation of each individual classifier's competence. Obviously, the most competitive classifier from the candidate classifiers might not be selected in the case of poor estimation, which could have negative effect on the performance of the classifier system. In contrast with the DCS system which selects the single best classifier for each unknown pattern, the DES system dynamically selects and combines an appropriate subset of classifiers from the candidate classifier pool based on the competence of individual classifier (Ko et al., 2008; Woźniak et al., 2014). In this way, DES-based classifier system takes advantage of both the selection and fusion approaches.

After selecting an appropriate subset of classifiers, an aggregation rule should be used to fuse the outputs obtained by the selected classifiers. There are many approaches proposed in the literature (Cruz, Sabourin, & Cavalcanti, 2018a) for aggregating the outputs of the selected classifiers. However, none of them consider the local distribution of examples in the region where the query sample is located. In this paper, we attempt to develop a new combination method with the aim to improve the performance of DES-based classification systems. With this objective, a distance-based weighting framework is proposed to dynamically adjust the hypothesis vector derived from the DES system. To do so, we use the average distance of k nearest neighbors of each class to adjust the hypothesis vector derived from the DES system for each query sample. With such consideration, the larger the distance is, the lower the weight is. This way, the decision boundary is more likely forced to move forward correct class label. One should note that our approach is different from the neighbors-based dynamic weighting method considered in dynamic classifier systems (i.e. DCW) (Sun, 2010) and the decomposition scheme (i.e. DRCW-OVO) (Galar, Fernández, Barrenechea, & Herrera, 2015). The main difference between DCW and DRCW-OVO with respect to our method is that our approach modifies the hypothesis vector depending on the distance measure instead of weighting the classifiers based on their competence.

In order to show the effectiveness and robustness of our approach, we present a thorough experimental study. Concretely, the candidate classifier pool is generated using both homogeneous and heterogeneous ensemble. A set of 30 real-world applications is used in our experiments. Accuracy and kappa (Cohen, 1960) are employed as the measure metrics to evaluate the performance of the algorithms. Finally, the proper statistical tests suggested in García, Fernández, Luengo, and Herrera (2010) and Derrac, García, Molina, and Herrera (2011) are used to study the significance of the results.

The remainder of this paper is organized as follows. Firstly, some related works with respect to this study are recalled in Section 2. Then, in Section 3, we present the details of our proposal. Next, Section 4 describes the experimental framework. The experimental study is carried out in Section 5 and the conclusions are presented in the final section.

2. Related works

In this section, we first describe the formulation of classification problems. Then, we introduce the procedure of DES-based methods. Finally, we recall some works on dynamic weighting.

2.1. Classification formulation

The classification approach consists of two stages: learning and prediction. In the first stage, a classification algorithm is employed to learn a classifier by using a given training dataset. In the second stage, a query sample is classified by employing the classifier derived from the first stage.

Let $T = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$ be a given training dataset, where $x_i \in \mathcal{X} \subseteq \mathcal{R}^f$ is the input feature vector and $y_i \in \mathcal{Y} = \{\omega_1, \omega_2, \dots, \omega_m\}$ is the class label with respect to the input feature vector x_i . In supervisor learning, it assumes that the unknown patterns are independent and identically distributed with the training dataset. The learning system aims to obtain a conditional probability $\hat{P}(Y|X)$ or a decision function $Y = \hat{f}(X)$, namely classifier φ , by using the classification learning algorithm H and the training dataset T . More specifically, for a given unknown pattern x_t , classifier φ produces a m -dimensional hypothesis vector $\mathbf{h}(x_t) = [h(\omega_1|x_t), h(\omega_2|x_t), \dots, h(\omega_m|x_t)]$, where $h(\omega_i|x_t)$ represents the support given to class ω_i for the query sample x_t . Without loss of generality, we can restrict $h(\omega_i|x_t) \in [0, 1]$ and $\sum_{i=1}^m h(\omega_i|x_t) = 1$. The decision of a classifier φ on the query sample x_t can be made as follows.

$$\varphi(x_t) = \arg \max_{i=1}^m h(\omega_i|x_t) \quad (1)$$

2.2. Dynamic ensemble selection

As different classifiers usually have different classification capabilities on different samples, the methods based on Dynamic Selection (DS) attempt to obtain the most appropriate classifier or ensemble of classifiers for a given unknown pattern. Among them,

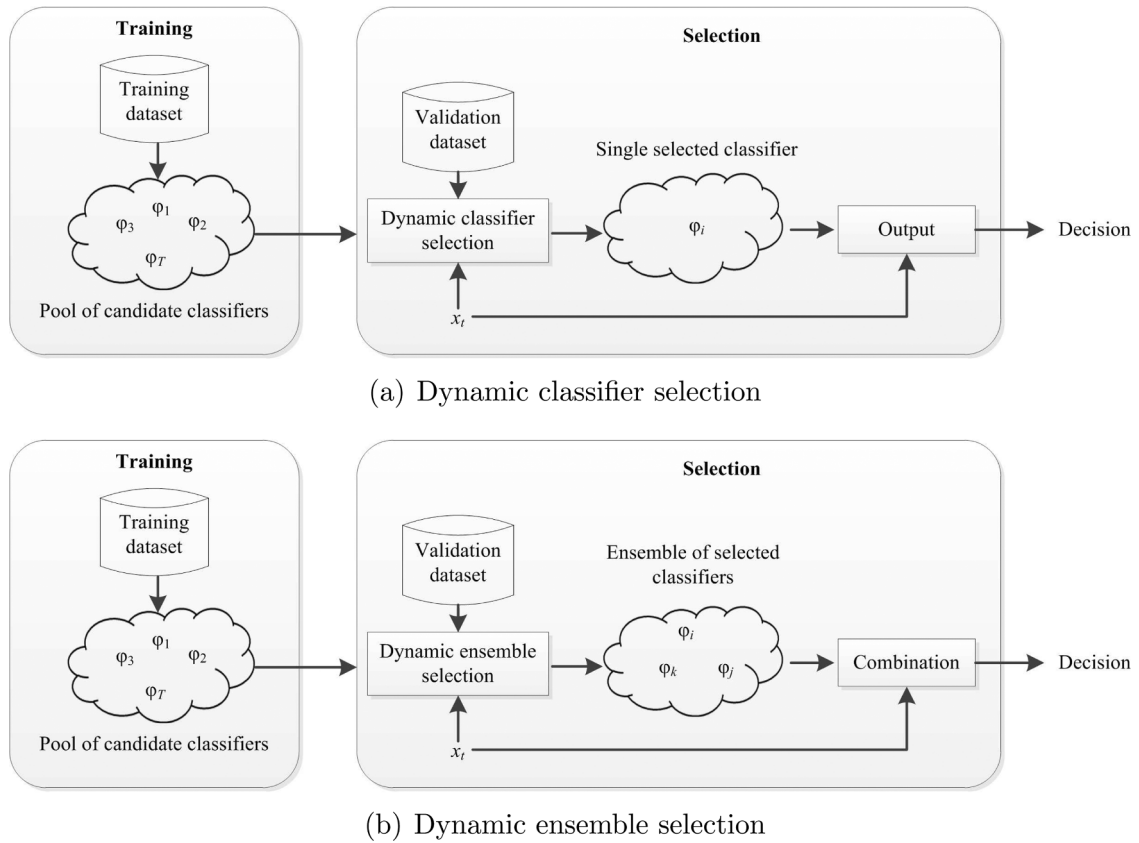


Fig. 1. Illustration of from dynamic classifier selection to dynamic ensemble selection.

DCS and DES are the two most popular strategies. The DCS strategy aims to select the best classifier for classifying a query sample. To accomplish this, the key component is the estimation of each individual classifier's competence. Many proposals for evaluating the competence have been developed (B., Sabourin, & Oliveira, 2014). Among them, accuracy-based measures are considered as one of the most popular techniques, such as the local accuracy (Woods et al., 1997), A Prior (Didaci, Giacinto, Roli, & Marcialis, 2005) and A Posterior (Giacinto & Roli, 1999) selection. Usually, the accuracy is obtained based on the local region defined by using k nearest neighbors surrounding the sample to be classified in the training dataset or the validation dataset in the feature space. An illustration of the DCS is shown in Fig. 1 (a).

Instead of selecting the most competitive single classifier from the candidate classifiers, the DES-based approaches aim to select the most suitable ensemble for each instance to be classified. Due to its simplicity to construct DES-based classifier systems, the DES approaches are widely applied in many real-world applications, such as credit scoring (Feng, Xiao, Zhong, Dong, & Qiu, 2019), fault detection (Amozegar & Khorasani, 2016), concept drift (Almeida, Oliveira, Britto, & Sabourin, 2018), and face recognition (Bashbaghi, Granger, Sabourin, & Bilodeau, 2017). Recently, dynamic selection methods are extended into many different contexts, including One-versus-One (OVO) decomposition strategies (Zhang, Luo, García, Tang, & Herrera, 2017; Zhang, Luo, Yu, Yuan, & Tang, 2018), one-class classification problems (Krawczyk, Galar, Woźniak, Bustince, & Herrera, 2018), imbalance learning tasks (García, Zhang, Altalhi, Alshomrani, & Herrera, 2018; Roy, Cruz, Sabourin, & Cavalcanti, 2018), and multi-label classification problems (Markatopoulou, Tsoumakas, & Vlahavas, 2015). Since in this paper we focus on improving the generalization performance of DES techniques, only these classic DES approaches are considered.

Generally, there are three main phases for constructing a DES system (Ko et al., 2008): (1) generation, where the training dataset is used to generate a candidate pool of classifiers; (2) selection, where an appropriate subset of classifiers are selected based on the competence of candidate classifiers in classifying a query sample; and (3) combination, where the selected classifiers are combined to produce the final decision. An illustration of the DES is shown in Fig. 1 (b).

2.2.1. Candidate classifier pool generation

The aim of the classifier pool generation stage is to construct a candidate classifier pool, $\Phi = \{\varphi_1, \varphi_2, \dots, \varphi_T\}$, containing T classifiers that are both accurate and diverse. The diversity of the base classifiers is the key component in the DES system, since it is impossible to improve the classification performance when combining classifiers with the same outputs. The way to generate diverse candidate classifiers can be roughly divided into two main groups: homogeneous ensemble and heterogeneous ensemble (Woloszynski, Kurzynski, Podsiadlo, & Stachowiak, 2012).

- **Homogeneous ensemble:** This method involves the usage of the same learning algorithm with different configurations to obtain the diversity in the candidate classifier pool. There are five main strategies to generate a pool of candidate classifiers using the same learning algorithm, including different initializations, different parameters, different architectures, different training datasets, and different feature sets.
- **Heterogeneous ensemble:** In this case, the base classifiers are generated by using different classification models, for example, Decision Trees (DT), Support Vector Machines (SVM), Neural Networks (NN) and k -Nearest Neighbors (k -NN). Different classification algorithms have different intrinsic properties to generate decision boundaries. Obviously, in this manner, the diversity is generated due to structurally different individual classifiers in the candidate classifier pool.

2.2.2. Ensemble selection

The selection stage is the main focus of the related works in the DES research committee. The goal of this stage is to select the most appropriate classifiers for each query sample needed to be classified. There are two main steps to accomplish the ensemble selection:

- **Definition of the region of competence:** The definition of a local region is to determine a small region in the feature space surrounding the query sample, x_t , where the level of competence of a candidate classifier is estimated. Many efforts have been devoted to improve the performance of DES systems via finding a better way to define these regions. Usually, the k -NN algorithm is used to define the region of competence, in which the k nearest neighbors of the query sample, x_t , are selected to obtain the local region (Ko et al., 2008). More recently, a local adaptive k -NN algorithm with prototype selection technique is applied to define the region of competence to improve the generalization performance of DES-based classifier systems in the overlap region (Cruz, Sabourin, & Cavalcanti, 2018b). Besides, clustering-based methods (Lin et al., 2014) are another effective strategy to define the region of competence, in which the competence of the candidate classifier is estimated in each cluster. Recently, a competence map using a potential function model (Woloszynski & Kurzynski, 2011; Woloszynski et al., 2012) and the usage of decision space (Cavalin, Sabourin, & Suen, 2012) are introduced to calculate the competence, respectively.
- **Selection criterion:** In the DES systems, the most important component is the selection criterion, which is used to measure the level of competence of the candidate classifiers in the region of competence. The strategies for the selection criterion can be divided into two categories (B. et al., 2014): individual-based and group-based measures. In the former, the individual performance of the candidate classifier is used to assess its competence level, including Ranking (Woods et al., 1997), Accuracy (Woods et al., 1997), Probabilistic (Woloszynski & Kurzynski, 2011), Behavior (Giacinto & Roli, 2001), Oracle (Ko et al., 2008), and Data complexity (Brun, Britto, Oliveira, Enembreck, & Sabourin, 2016). The latter presents the measures where the interaction between the classifiers in the candidate pool is taken into account. In this respect, Diversity (Santos, Sabourin, & Maupin, 2009), Data Handling (Xiao, Xie, He, & Jiang, 2012) and Ambiguity (Santos, Sabourin, & Maupin, 2007) can be included in this category. However, Cruz, Sabourin, Cavalcanti, and Ren (2015) argue that using only one selection criterion is not sufficient to evaluate the performance of a candidate classifier. In order to simultaneously consider different selection criteria, several works, namely META-DES (Cruz et al., 2015) and META-DES-Oracle (Cruz, Sabourin, & Cavalcanti, 2017), are proposed to handle the DES problem using meta-learning, in which the meta-features of the meta-problem are different criteria used to measure the level of competence of the base classifier. Instead of simply selecting the classifiers with the best competence under a certain criterion, a greedy randomized selection of appropriate classifiers to form the ensemble is proposed (Ye & Dai, 2018). With the aim to handle these query samples located in the indecision region, Oliveira, Cavalcanti, and Sabourin (2017) propose a new dynamic selection method named online pruning of base classifiers, in which these classifiers with decision boundaries crossing the region of competence are pre-selected. More recently, an enhanced FIRE-DES named FIRE-DES++ that removes noise and reduces the overlap of classes in the validation dataset is developed, in which the region of competence is defined as an equal number of samples of each class (Cruz, Oliveira, Cavalcanti, & Sabourin, 2019).

2.2.3. Classifier combination

The combination phase consists in fusing the outputs obtained by the selected classifiers according to a predefined aggregation rule. There are two main strategies for the combination phase:

- **Heuristic method:** In these methods, a non-trainable combination rule is developed to fuse the outputs of the selected classifiers. Several such kind of rules have been proposed, including Sum, Product, Maximum, Minimum, Median and Majority Voting schemes (Kittler, Hatef, Duin, & Matas, 1998), Borda count (Ho, Hull, & Srihari, 1994), Behavior Knowledge Space (Huang & Suen, 1995), Decision Templates (Kuncheva, Bezdek, & Duin, 2001) and Dempster-Shafer combination (Kessentini, Burger, & Paquet, 2015; Lu, 1996). The performance of different combination rules can be found in the literature (Duin, 2002).
- **Trainable approach:** Different from the heuristic method that uses a fixed combination rule, the outputs of the selected classifiers in this strategy are considered as the inputs for another learning algorithm, which aims to draw the function for combining the selected classifiers. Several studies have demonstrated the fact that trainable approaches perform better than heuristic rules. A well-known trainable combination method is using neural networks to aggregate the outputs of the base classifiers (Rahman & Verma, 2013). Another interesting trainable combination method is the Mixture of Experts (ME) (Jacobs, Jordan, Nowlan, & Hinton, 1991), where the base classifiers and the combination function are trained together.

2.3. Dynamic weighting methods

Dynamic classifier weighting (DCW) (Sun, 2010) is a kind of approach which is closely related to DCS and DES. Instead of selecting the most appropriate classifiers (only one in DCS), DCW methods combine the classifiers by weighted voting depending on the competence of the classifiers. Similar to DES methods, dynamic weighting strategies use the region of competence surrounding the query sample x_t . However, instead of selecting and combining a group of classifiers with high competence level, dynamic weighting methods aggregate the outputs of all classifiers in which the most competent classifier receives a higher voting weights. That is, the more competent the classifier is, the higher weights will be assigned to it.

Several works have been developed for the dynamic weighting strategy. examples of such strategies are the local classifier weighting by quadratic programming (Cevikalp & Polikar, 2008), the dynamic integration of classifiers (Tsybal, Pechenizkiy, Cunningham, & Puuronen, 2008), and the fuzzy dynamic classifier aggregation (Štefka & Holeňa, 2015). In addition, a hybrid of ensemble selection and dynamic weighting is also available for constructing a dynamic classification system (Vriesmann, Britto, Oliveira, Koerich, & Sabourin, 2015). In these approaches, the most competent classifiers are selected using a certain assessment of competence, and then the dynamic weighting scheme is used to combine the selected classifiers. Experimental results conducted in Cruz et al. (2018a) indicate that the hybrid approaches usually outperform the pure dynamic weighting methods.

3. The proposed methodology

In this section, we present the details of the proposed method for empowering the DES-based methods. In Section 3.1 we introduce our framework for enhancing the classification ability of DES methods. In Section 3.2 we describe the well-known DES approaches employed in this paper to obtain the ensemble. In Section 3.3 confidence estimation for the selected ensemble is presented, while in Section 3.4 we explain the way how to measure the distance in our method.

3.1. Distance-based weighting method for dynamic ensemble selection

Most of current works on DES-based methods mentioned in Section 2.2 focus on how to select a group of classifiers from the candidate classifier pool using the local information surrounding the query sample (Cruz et al., 2018a) and (B. et al., 2014). However, the previous studies (Kessentini et al., 2015; Rahman & Verma, 2013) do not take into account the local information when combining the selected classifiers, which might hinder the classification performance of the DES classifier system. Considering this fact, we in this paper propose a distance-based weighting framework in order to improve the classification ability of the DES-based classification systems.

In our proposed framework, once a group of classifiers for a query sample x_t has been selected by a given DES method, the operating procedure of our proposal is as follows:

- step 1 Compute the confidence estimation for each class using the extent of the consensus of the selected ensemble and store the hypothesis vector $\mathbf{h}(x_t) = [h(\omega_1|x_t), h(\omega_2|x_t), \dots, h(\omega_m|x_t)]$.
- step 2 Select the k nearest neighbors of each class for the query sample x_t and store the average distance of k nearest neighbors of each class in a vector $\mathbf{d} = [d_1, d_2, \dots, d_m]$.
- step 3 A new hypothesis vector $\mathbf{h}^{new}(x_t)$ is created where the element $h^{new}(\omega_i|x_t)$ are weighted as follows:

$$h^{new}(\omega_i|x_t) = h(\omega_i|x_t) \times w_i \quad (2)$$

where w_i is the adjustment coefficient computed as

$$w_i = \frac{d_i^2}{\sum_{j=1}^m d_j^2} \quad (3)$$

where d_i is the average distance of the query sample x_t to the k nearest neighbors of class ω_i .

- step 4 Use the following equation to obtain the final class.

$$\varphi(x_t) = \arg \max_{i=1}^m h^{new}(\omega_i|x_t) \quad (4)$$

As mentioned above, the key components of our method include: the selection of an appropriate group of classifiers, the measurement of confidence estimation for an ensemble, and the calculation of the adjustment coefficient, which are described in the following sections.

3.2. Dynamic ensemble selection techniques

In the proposed method, we first selects an ensemble from a candidate classifier pool for each query sample. To do so, a procedure for selecting a group of classifiers should be employed in our method. On this account, we in this paper consider six well-known DES techniques to select an appropriate group of classifiers for each query sample. The details of the selected DES methods are described

in the following subsections.

3.2.1. KNORA-Eliminate

The K-Nearest-ORAcles Eliminate (KNORA-E) [Ko et al. \(2008\)](#) system selects a subset of classifiers that correctly classify all the k nearest neighbors surrounding the query sample x_t . If there is no classifier can fit this criterion, i.e. no classifier can correctly classify all the k nearest neighbors of the query sample, then the value of k is decreased until at least one classifier is selected. The selected classifiers are combined by the majority voting rule and the value of $k = 10$ is used.

3.2.2. KNORA-Union

In contrast to the KNORA-E system, the K-Nearest-ORAcles Union (KNORA-U) [\(Ko et al., 2008\)](#) system classifies the query sample x_t using a majority voting rule where a classifier correctly classifies the j -nearest neighbor should submit a vote on x_t . One should note that, if more than one nearest neighbor is considered, a classifier can have more than one vote, since it might correctly classify more than one nearest neighbor. The value of $k = 10$ is used.

3.2.3. Dynamic ensemble selection performance

The DES-Performance (DES-P) [\(Woloszynski et al., 2012\)](#) system selects a subset of classifiers for each query sample x_t using the measure of competence based on random classification. The classifiers with the performance in a weighted k -nearest neighborhood of the query sample x_t greater than the random classifier are selected from the candidate ensemble, i.e.

$$C(\varphi|x_t) = \text{Acc}(\varphi|x_t) - \frac{1}{m} \quad (5)$$

where $\text{Acc}(\varphi|x_t)$ estimates the classification accuracy of the classifier φ on the weighted k nearest neighbors of the instance x_t to be classified in the validation dataset. The value of $k = 10$ is used for which the best results were obtained in the DES-P system. $\frac{1}{m}$ (m denotes the number of class labels.) indicates the performance of a random classifier.

3.2.4. Kullback–Leibler divergence

In the DES Kullback–Leibler (DES-KL) [Woloszynski et al. \(2012\)](#) system, the information theory is used to estimate the competence of candidate classifiers. In order to measure the distance between the values of discriminant functions and the probability of random classification, the Kullback–Leibler (KL) divergence between the vector of discriminant functions $[h(\omega_1|x), h(\omega_2|x), \dots, h(\omega_m|x)]$ produced by the classifier φ and the uniform distribution $[\frac{1}{m}, \dots, \frac{1}{m}]$ is used to calculate the source competence $c(\varphi|x)$ for each example x in the validation dataset. The competence of the classifier φ for the query sample x_t is obtained by a weighted sum of the source competence, whose sign is given by the expression $h(\omega_i|x) - \max_{j=1, \dots, m; j \neq i} h(\omega_j|x)$, where i denotes the index of the real class for the example x . Then, the competence of the classifier φ for the query sample x_t is obtained by a weighted sum of the source competences as follows.

$$C(\varphi|x_t) = \sum_{x \in V} c(\varphi|x) \exp(-\text{dist}(x_t, x)^2) \quad (6)$$

where $\text{dist}(x_t, x)$ is the distance between the query sample x and example x_t . The classifiers with nonnegative competences are selected to construct the DES-KL system.

3.2.5. DES-competence based on label outputs and weighted majority voting

The DES-competence based on label outputs and weighted majority voting (DES-CV) [\(Woloszynski & Kurzynski, 2011\)](#) firstly constructs the source competence function $c(\varphi|x)$ for each classifier in the candidate classifier pool using a probabilistic model based on the Randomized Reference Classifier (RRC), i.e. $c(\varphi|x) = P_c(RRC|x)$, where $P_c(RRC|x)$ is the conditional probability of correct classification for RRC [\(Berger, 2013\)](#). Then, a Gaussian potential function with the distance is used for obtaining the potential function model as follows.

$$C(\varphi_l|x_t) = \frac{\sum_{j=1}^n c(\varphi_l|x_j) \exp(-\text{dist}(x_t, x_j)^2)}{\sum_{j=1}^n \exp(-\text{dist}(x_t, x_j)^2)} \quad (7)$$

where x_t is the sample needed to be classified and x_j denotes each example in the validation dataset.

The classifiers with the competence greater than the probability of random classification is selected from the candidate ensemble for each query sample x_t , i.e. $C(\varphi_l|x_t) > \frac{1}{m}$, $l = 1, \dots, L$, $L \leq T$. Finally, the selected classifiers are combined by using the weighted majority voting rule, where the competence is considered as the weight for each classifier.

$$h^{CV}(\omega_i|x_t) = \sum_{l=1}^L C(\varphi_l|x_t) I(h_l(\omega_i|x_t) = \max_{k \in \mathcal{Y}} h_l(\omega_k|x_t)) \quad (8)$$

where $I(\bullet)$ is the indicator function, in which the value is equal to 1 (0) if the statement is true (false). The final result of the DES-CV system is obtained as follow:

$$\varphi_{DES-CV}(x_t) = \arg \max_{i=1}^m h^{CV}(\omega_i | x_t) \quad (9)$$

3.2.6. DES-competence based on continue-valued outputs and weighted class supports

In the method of DES-competence based on continue-valued outputs and weighted class supports (DES-CS) (Woloszynski & Kurzynski, 2011), the competence function and selection operation is the same as that in the DES-CV system, except that the DES-CS system uses a weighted vector of class supports instead of the weighted majority voting rule.

$$h^{CS}(\omega_i | x_t) = \sum_{l=1}^L C(\varphi_l | x_t) h_l(\omega_i | x_t) \quad (10)$$

Finally, the output class label of DES-CS system for the query sample x_t is:

$$\varphi_{DES-CS}(x_t) = \arg \max_{i=1}^m h^{CS}(\omega_i | x_t) \quad (11)$$

3.3. Confidence estimation for the selected ensemble

In our proposed method, the confidence of the selected ensemble should be estimated in order to obtain the initial hypothesis vector if it is not produced by the DES method. Some works (Hansen, Liisberg, & Salamon, 1997; Santos et al., 2007; 2008) have shown that it is reasonable to calculate the confidence of an ensemble decision by measuring the extent of the consensus associated with it. That is, the higher the consensus among classifiers in the ensemble, the higher confidence in the decision. Considering a classification problem with class labels $\mathcal{Y} = \{\omega_1, \omega_2, \dots, \omega_m\}$, the confidence is related to the posterior probability $P(\omega_i | x_t)$ that is the degree of x_t belonging to ω_i . Given an ensemble of classifiers $EOC = \{\varphi_1, \varphi_2, \dots, \varphi_L\}$ and the output class label of the k_{th} classifier $O_k(x_t)$. The number of votes $v(\omega_i | x_t)$ for class ω_i given x_t is calculated as follows:

$$v(\omega_i | x_t) = |\varphi_k: O_k(x_t) = \omega_i| \quad (12)$$

Then, the consensus decision is obtained as follows:

$$mv(x_t) = \arg \max_{i=1}^m v(\omega_i | x_t) \quad (13)$$

Finally, the support for class ω_i using the extent of the consensus on the example x_t is

$$h(\omega_i | x_t) = P(\omega_i | x_t) = \frac{v(mv(x_t) | x_t)}{|EOC|} \quad (14)$$

3.4. Distance measure metric

In our method, a distance function should be used to calculate the adjustment coefficient. Although there have been many distance functions proposed, by far the most commonly used is the Euclidean distance function which is developed for the numerical attributes. On the other hand, the Value Difference Metric (VDM) (Stanfill & Waltz, 1986) was developed to define an appropriate distance function for the nominal attributes. However, in many real-world applications, the measure of distance involves in both numerical and nominal attributes. To overcome this issue, a distance function named Heterogeneous Value Difference Metric (HVDM) (Wilson & Martinez, 1996) that combines Euclidean distance and VDM to simultaneously handle these continuous and nominal attributes. Therefore, in this study the HVDM is employed to measure the distance between examples, which is also successfully used in many algorithms (Galar et al., 2015; Zhang et al., 2017). In the following, we briefly describe the HVDM method.

In the HVDM, it calculates the distance between two input examples x_1 and x_2 as the following equation:

$$d_{HVDM}(x_1, x_2) = \sqrt{\sum_{a=1}^f dist_a^2(x_1, x_2)} \quad (15)$$

where f stands for the number of attributes and the function $dist_a(x_1, x_2)$ defined as in Eq. (16) represents the distance metric between two values x_1 and x_2 for attribute a :

$$dist_a(x_1, x_2) = \begin{cases} 1, & \text{if } x_1 \text{ or } x_2 \text{ is unknown; otherwise...} \\ normalized_vdm_a(x_1, x_2), & \text{if } a \text{ is nominal} \\ normalized_diff_a(x_1, x_2), & \text{if } a \text{ is continuous} \end{cases} \quad (16)$$

Hence, $dist_a(x_1, x_2)$ employs one of these two functions to calculate the distance between values of attributes according to the type of attribute a . The function measures the distance for nominal attributes is defined as follows:

$$normalized_vdm_a(x_1, x_2) = \sqrt{\sum_{i=1}^m \left| \frac{n_{a,x_1,\omega_i}}{n_{a,x_1}} - \frac{n_{a,x_2,\omega_i}}{n_{a,x_2}} \right|} \quad (17)$$

where m represents the number of classes, n_{a,x_1} stands for the number of examples in the training dataset with value x_1 for attribute a , and n_{a,x_1,ω_i} is the number of examples in the training dataset with value x_1 for attribute a which is labeled as class ω_i .

When calculating the distance for continuous attributes, the following equation is used:

$$\text{normalized_diff}_a(x_1, x_2) = \frac{|x_1 - x_2|}{4 \cdot \sigma_a} \quad (18)$$

where σ_a stands for the standard deviation of the values of attribute a in the training dataset.

4. Experimental framework

In this section, we first in Section 4.1 present the base learners and corresponding parameter configurations used for generating candidate classifiers. Then, the description of datasets selected for the experiments is presented in Section 4.2. Next, the assessment metrics used to evaluate the performance of methods are explained in Section 4.3. Finally, in Section 4.4, we present the statistical tests adopted to analyze the performance of tested methods.

4.1. Base learners and parameters configurations

In this study, we consider the MATLAB platform using PR-Tools developed by Duin et al. (2007) to carry out the experimental study. In our experiments, we use both homogeneous and heterogeneous learning algorithms to generate the candidate classifier pool. The homogeneous classifier pool consists of forty decision trees with the Gini splitting criterion, the pruning is used, and impure nodes must have 10 or more examples to be split. The diversity among individual classifiers in the homogeneous ensemble is obtained by resampling with replace like used in Bagging. With respect to the heterogeneous ensemble, the description of the candidate learning algorithms and the corresponding parameter configurations are presented in Table 1. In this scenario, the appropriate ensemble is selected from a classifier pool consisting of forty different base classifiers. This way, it is clear that structurally different individual classifiers supply the diversity when using heterogeneous classifiers.

4.2. Datasets

In this paper, in order to demonstrate the usefulness and effectiveness of our method, 28 datasets from the UCI repository¹ and 2 from the Ludmila Kuncheva Collection (LKC) of real medical data² were selected to analyze the algorithms. The properties of the datasets are shown in Table 2. For each dataset, it includes the number of examples (#Ex.), the number of attributes (#Atts.), number of numerical (#Num) and nominal (#Nom) attributes, number of classes (#Cl.), and distribution of class (#Dc). Instances with missing values in some datasets were removed before doing the partitions.

The average results are obtained by means of 5-fold cross-validation (Moreno-Torres, Saez, & Herrera, 2012). That is, each dataset is split into 5 folds and each fold contains 20% of the instances of the dataset. For each fold, the algorithm is trained with the instances contained in the remaining folds (80% of the instances of the dataset) and then tested by the current fold.

4.3. Assessment of the classification performance

In this study, classification rate (also known as accuracy rate) and Cohen's kappa (Cohen, 1960) are considered as the assessment metric to evaluate the performance of the tested methods due to their simplicity and successful application when classification problems are treated.

Accuracy rate has been by far the most widely used metric for evaluating the performance of classifiers for years. It is computed as the ratio of the number of instances correctly classified (successful hits) to the total number of instances.

In the case of classification problems, it might not show the real behavior of the tested methods by only considering accuracy rate. For this reason, we in this study employ Cohen's kappa as another measure to evaluate the performance of methods, since it compensates for random hits. In order to compute such measure clearly, the confusion matrix of results in a classification task is presented in Table 3. From the confusion matrix, Cohen's kappa is calculated as follows.

$$\text{kappa} = \frac{n \sum_{i=1}^m u_{ii} - \sum_{i=1}^m n_{ri} n_{ci}}{n^2 - \sum_{i=1}^m n_{ri} n_{ci}} \quad (19)$$

where n is the number of total instances, m is the number of classes in the problem, u_{ii} stands for the number of true positives for each class, n_{ri} ($n_{ri} = \sum_{j=1}^m u_{ij}$) and n_{ci} ($n_{ci} = \sum_{j=1}^m u_{ji}$) denotes the i -th row's and column's counts, respectively. It is easy to find that the value of Cohen's kappa ranges from -1 to 1 .

¹ <http://archive.ics.uci.edu/ml/datasets.html>.

² http://pages.bangor.ac.uk/~mas00a/activities/real_data.htm.

Table 1

Description of parameters configuration for the individual classifiers.

No.	Base learner	Description
1, 2, 3, 4 and 5	TREE	Decision tree classifier with the Gini splitting criterion and the pruning level set to 0, 2, 4, 8 and 10.
6, 7, 8, 9 and 10	TREE	Decision tree classifier with the information gain criterion and the pruning level set to 0, 2, 4, 8 and 10.
11, 12, 13, 14, 15 and 16	k-NN	k-nearest neighbors classifier with $k = 1, 3, 5, 7, 10, 15$.
17	RNNC	Random neural net classifier with 20 neurons in each hidden layer
18, 19 and 20	BPNN	Feed-forward back-propagation neural network classifier containing 1, 1 and 2 hidden layers with 10, 5 and 5 neurons in each hidden layer and the number of epochs set to 200, 1000 and 100, respectively.
21, 22 and 23	RBF	Radial basis function neural network and the number of RBF units in hidden layers set to 5, 10 and 20.
24, 25, 26, 27, 28, 29, 30 and 31	SVM	Support vector machine classifier with the polynomial, radial basis, sigmoid and distance kernels and regularization parameter $C=1$ and 5 for each kernel.
32 and 33	LMNC	Levenberg-Marquardt feed-forward neural network classifier containing 1 and 2 hidden layers with 10 and 5 neurons in each hidden layer and the number of epochs set to 1000 and 200, respectively.
34	NEURC	Automatic Levenberg-Marquardt feed-forward neural network classifier containing one hidden layer.
35	NAIVEB	Naïve Bayes classifier.
36	PARZENC	Parzen classifier with the Gaussian kernel.
37	PARZENDC	Parzen density classifier with the Gaussian kernel.
38	QDC	Quadratic discriminant classifiers based on normal distributions with different covariance matrix for each class.
39	LDC	Linear discriminant classifiers based on normal distributions with the same covariance matrix for each class.
40	UDC	Uncorrelated normal densities based quadratic Bayes classifier.

Table 2

Summary description of the datasets used in the experiments.

ID	Dataset	#Ex.	#Atts.	#Num.	#Nom.	#Cl.	#Dc.	Source
Aus	Australian	690	14	8	6	2	307/383	UCI
Aut	Automobile	159	25	15	10	6	46/29/13/48/20/3	UCI
Bal	Balance	625	4	4	0	3	288/49/288	UCI
Bre	Breast-Tissue	106	9	9	0	6	22/21/14/15/16/18	UCI
Car	Car	1728	6	0	6	4	384/69/1210/65	UCI
Cle	Cleveland	297	13	5	8	5	160/54/35/35/13	UCI
Der	Dermatology	358	34	1	33	6	111/60/71/48/48/20	UCI
Eco	Ecoli	336	7	7	0	8	143/77/2/2/35/20/5/52	UCI
For	Forest	523	27	27	0	4	159/86/83/195	UCI
Ger	German	1000	20	3	17	2	700/300	UCI
Gla	Glass	214	9	9	0	6	70/76/17/29/13/9	UCI
Ilp	Indian-liver-patient	579	10	9	1	2	414/165	UCI
Ion	Ionosphere	351	33	32	1	2	126/225	UCI
Iri	Iris	150	4	4	0	3	50/50/50	UCI
Lar	laryngeal3	353	16	16	0	3	53/218/82	LKC
Lym	Lymphography	148	18	3	15	4	4/61/81/2	UCI
New	New-thyroid	215	5	5	0	3	150/35/30	UCI
Pen	Penbased	1100	16	16	0	10	115/114/114/106/114/106/105/115/105/106	UCI
Pim	Pima	768	8	8	0	2	268/500	UCI
Son	Sonar	208	60	60	0	2	97/111	UCI
Use	User-modeling	403	5	5	0	4	102/129/122/50	UCI
Ver	Vertebral-column	310	6	6	0	3	60/150/100	UCI
Voi	Voice3	238	10	9	1	3	52/18/168	LKC
Vow	Vowel	990	13	13	0	11	90/90/90/90/90/90/90/90/90/90	UCI
Wdb	Wdbc	569	30	30	0	2	357/212	UCI
Win	Wine	178	13	13	0	3	59/71/48	UCI
Wis	Wisconsin	683	9	9	0	2	444/239	UCI
Wqr	Wine-Quality-Red	1599	11	11	0	6	10/53/681/638/199/18	UCI
Yea	Yeast	1484	8	8	0	10	463/5/35/44/51/163/244/429/20/30	UCI
Zoo	Zoo	101	16	0	16	7	41/20/5/13/4/8/10	UCI

4.4. Statistical tests

It is necessary to employ proper statistical tests for analyzing the obtained experimental results. In this paper, we adopt non-parametric hypothesis testing techniques recommended in the literature (Derrac et al., 2011; García et al., 2010) where a set of robust and simple non-parametric tests for statistical comparison of classification algorithms is presented. Specially, we carry out pairwise comparison of the new DES method using our proposed weighting framework with respect to the original version for each selected DES method. For this reason, Wilcoxon signed-rank test (Wilcoxon, 1945) is employed as non-parametric statistical procedure to

Table 3
Confusion matrix for an m -class problem.

Actual class	Output class				
	ω_1	ω_2	...	ω_m	Total
ω_1	u_{11}	u_{12}	...	u_{1m}	n_{r1}
ω_2	u_{21}	u_{22}	...	u_{2m}	n_{r2}
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
ω_m	u_{m1}	u_{m2}	...	u_{mm}	n_{rm}
Total	n_{c1}	n_{c2}	...	n_{cm}	n

perform comparison between our method and the other. The employed statistical test allows us to check whether the proposed framework is useful for DES systems or not.

5. Results and discussion

The aim of this section is to verify the effectiveness and robustness of the proposed framework for improving the performance of DES-based systems. With this objective, we first carry out a pairwise comparison between the proposed framework and the original DES version considering homogenous ensemble and heterogenous ensemble in Sections 5.1 and 5.2, respectively. With respect to the DES method used in the experiments, we employ the six well-known DES approaches which are described in the previous section, i.e. KNORA-E, KNORA-U, DES-P, DES-KL, DES-CV and DES-CS. Finally, we perform the comparison of the proposed method and the well-known ensemble methods in Section 5.3.

5.1. Homogeneous ensemble

The complete results obtained from the homogeneous ensemble are shown in Table 4 and Table 6, with classification accuracy and kappa, respectively. The best result in each dataset for each performance measure is highlighted in **bold-face**. In general, the

Table 4
Results of classification accuracy obtained for using homogeneous ensemble. The best result in each dataset for each DES method is highlighted in **bold-face**.

Dataset	KNORA-E		KNORA-U		DES-P		DES-KL		DES-CV		DES-CS	
	Orig.	DW	Orig.	DW	Orig.	DW	Orig.	DW	Orig.	DW	Orig.	DW
Aus	85.22	85.94	86.09	86.23	85.22	85.36	85.36	85.22	85.36	85.22	85.65	86.52
Aut	86.19	87.44	84.33	84.98	83.71	84.33	83.71	84.98	83.71	85.60	85.58	85.58
Bal	78.56	79.20	85.12	86.24	85.44	87.04	85.76	87.52	85.76	87.52	85.60	86.24
Bre	64.13	65.00	69.76	70.69	68.81	70.69	66.94	69.74	65.89	69.74	66.04	70.64
Car	97.40	97.63	96.88	96.64	96.70	96.64	96.30	96.64	96.59	96.59	96.12	96.59
Cle	52.47	52.46	55.79	57.48	54.84	54.49	56.80	57.15	56.47	57.15	56.80	56.48
Der	94.68	95.25	95.24	96.92	94.96	96.65	94.96	96.37	95.81	97.20	94.96	96.92
Eco	77.70	79.18	80.37	83.05	80.37	83.05	79.78	82.16	80.37	83.05	79.47	81.55
For	87.77	89.10	87.57	87.76	87.57	87.76	87.57	87.76	87.95	88.33	88.14	88.52
Ger	74.20	74.20	79.20	78.10	76.50	76.80	77.00	76.80	78.60	78.40	77.70	76.30
Gla	75.69	77.09	74.31	76.17	74.32	75.70	73.39	74.77	73.39	74.77	74.32	75.24
Ilp	72.03	73.23	71.16	71.85	72.03	72.54	70.82	72.37	70.13	72.37	70.99	72.89
Ion	92.03	93.46	92.03	92.89	91.74	92.89	92.31	93.74	91.74	92.89	91.74	93.74
Iri	93.33	94.00	95.33	96.00	96.00	96.00	95.33	96.00	95.33	96.00	94.67	96.00
Lar	68.24	69.94	73.36	74.20	72.22	73.91	72.50	74.20	72.51	73.92	73.07	74.48
Lym	78.94	82.34	79.08	83.08	77.08	83.79	77.08	83.79	79.79	83.08	79.77	83.77
New	95.35	96.28	94.88	96.28	93.95	95.35	93.95	95.35	94.42	95.35	94.42	95.35
Pen	94.45	96.45	93.27	96.45	93.91	96.73	92.82	96.36	93.64	96.73	93.91	96.73
Pim	72.65	73.44	74.61	74.87	75.00	75.39	75.52	76.30	75.13	75.26	75.65	76.04
Son	84.62	86.09	80.78	85.57	80.31	83.68	79.83	85.57	77.89	82.24	78.39	83.67
Use	90.30	91.53	92.02	93.52	92.27	93.27	92.27	93.27	93.02	92.52	93.27	93.27
Ver	84.84	84.19	82.90	84.84	83.23	84.52	80.65	83.87	80.00	83.55	82.90	84.84
Voi	73.14	75.24	76.05	77.31	76.44	77.30	76.44	77.73	76.89	77.73	78.56	79.00
Vow	91.62	94.65	89.80	96.26	88.59	96.06	90.20	96.67	90.10	96.57	90.30	96.57
Wdb	94.56	95.61	95.26	96.13	95.43	96.13	95.26	96.13	95.61	96.84	95.26	96.31
Win	97.16	97.73	97.71	97.73	95.43	96.60	98.29	97.73	96.57	97.73	97.14	97.73
Wis	96.79	97.22	96.50	96.79	96.50	96.93	96.35	96.79	96.64	96.93	96.64	96.78
Wqr	64.86	65.98	67.04	67.42	68.85	69.17	65.92	67.42	67.11	67.29	66.23	68.29
Yea	51.15	53.30	58.56	59.44	58.96	59.70	59.10	59.57	59.03	59.57	58.96	59.64
Zoo	94.05	95.05	91.10	92.10	93.05	94.05	91.10	92.10	92.10	92.10	89.10	96.00
Avg.	82.14	83.27	83.20	84.57	82.98	84.42	82.78	84.47	82.92	84.41	83.04	84.72

Table 5

Results of Wilcoxon signed-rank test for the comparison between the proposed approach and the original method when homogeneous ensemble is used (classification accuracy).

Comparison	R^+	R^-	Hypothesis	p -value
KNORA-E-DW vs. KNORA-E-Orig.	454.50	10.50	Rejected for KNORA-E-DW at 5%	0.000006**
KNORA-U-DW vs. KNORA-U-Orig.	444.00	21.00	Rejected for KNORA-U-DW at 5%	0.000014**
DES-P-DW vs. DES-P-Orig.	455.50	9.50	Rejected for DES-P-DW at 5%	0.000005**
DES-KL-DW vs. DES-KL-Orig.	453.00	12.00	Rejected for DES-KL-DW at 5%	0.000006**
DES-CV-DW vs. DES-CV-Orig.	444.50	20.50	Rejected for DES-CV-DW at 5%	0.000015**
DES-CS-DW vs. DES-CS-Orig.	442.50	22.50	Rejected for DES-CS-DW at 5%	0.000023**

** Close to the p -value means that statistical differences are found with $\alpha = 0.05$ (95% confidence).

proposed weighting framework for DES-based systems is remarkable for both performance measure metrics, since the average performance of the proposed method is better for each tested DES method and the proposed dynamic weighting method outperforms the original DES approaches in most tested datasets. Concretely, (1) with respect to classification accuracy (Table 4), the proposed weighting framework outperforms these original DES-based methods by 1.14%, 1.36%, 1.44%, 1.69%, 1.49% and 1.68%, respectively. (2) Considering the performance measure of kappa (Table 6), the proposed weighting framework outperforms the original methods by 0.018462, 0.020067, 0.022123, 0.027519, 0.023237 and 0.025074, respectively. However, we cannot draw meaningful conclusions without using proper statistical tests. Therefore, we carry out the Wilcoxon signed-rank tests for the pairwise comparison between the weighting framework and the original DES version.

The statistical results for the homogeneous ensemble are presented in Tables 5 and 7, in which R^+ stands for the sum of the proposed weighting method and R^- for the original ones, for both classification accuracy and kappa performance measures, respectively. Looking at these statistical results, one can easily find the superiority of the proposed weighting framework, because all the hypothesis of equivalence for both performance measures are rejected. That is, the proposed weighting framework can significantly improve the performance of DES systems when using the homogeneous ensemble to generate candidate classifiers.

5.2. Heterogeneous ensemble

For the heterogeneous ensemble, we present the whole results for classification accuracy and kappa in Tables 8 and 10, respectively, in which the best result in each dataset for each performance measure is also highlighted in **bold-face**. Although the average performance of the proposed method is remarkable, the number of datasets in which the proposed weighting method performs better is not as remarkable as in the case of homogeneous ensemble. Regarding KNORA-E, DES-KL and DES-CS, we can see that the proposed method achieves results that are either superior or equivalent to the original version in most of datasets, while the results are almost tied in the cases of KNORA-U, DES-P and DES-CV. The relatively worse performance of the heterogeneous ensemble with respect to the homogeneous ensemble could be that the selected heterogeneous ensemble is chaotic and is more likely to generate inconsistency.

In order to extract meaningful conclusions, we have to employ the proper statistical methods to analyze the experimental results. To do so, we employ the same statistical tests as used in the previous section to carry out the pairwise comparison to check in the scenario of the heterogeneous ensemble whether the proposed method is significant better than the original version or not.

The results of the statistical tests when using heterogeneous candidate classifier pool are presented in Tables 9 and 11, where R^+ presents the sum of the ranks for the proposed method and R^- for the original DES method, for both classification accuracy and kappa performance measures, respectively. Regarding the results of statistical tests, there are statistical differences can be found in the cases of KNORA-E, DES-KL and DES-CS, while the Wilcoxon signed-rank test does not reject the equivalence between the methods using either accuracy measure or kappa measure in the scenarios of KNORA-U, DES-P and DES-CV. Despite no statistical differences being found in the later cases, we can still highlight the robust performance of the proposed dynamic weighting framework for DES-based techniques, as in these cases the values of R^+ are considerably greater than that of R^- . Above all, analyzing these results, we can easily find that the observed statistical results support the usefulness and effectiveness of our novel weighting framework for the DES systems when using heterogeneous classifiers to generate the candidate classifier pool.

Comparing these results obtained in different scenarios of homogeneous ensemble and heterogeneous ensemble, we can find that the way to generate the candidate classifiers in our method significantly impacts the observed results, both classification accuracy and kappa. The obtained improvement in the scenario of the heterogeneous ensemble is slightly lower than the homogeneous ensemble. Anyway, based on the obtained results, it allows us to highlight the superiority of the proposed weighting procedure for DES systems when compared with the original version.

5.3. Comparison with the state-of-the-art approaches

In previous sections, we have shown the effectiveness of our proposed method with respect to the original version, the comparison of the proposed method and the traditional ensemble methods is carried out in this section. Considering this objective, we select Bagging (Breiman, 1996), AdaBoost (Freund & Schapire, 1996), and Random Forest (RF) (Breiman, 2001) as benchmark methods, DES-CS with the homogeneous ensemble as the representative for the proposed method (abbreviated as DES-DW). The whole results

Table 6
Average kappa results obtained for using homogeneous ensemble. The best result in each dataset for each DES method is highlighted in **bold-face**.

Dataset	KNORA-E		KNORA-U		DES-P		DES-KL		DES-CV		DES-CS	
	Orig.	DW	Orig.	DW	Orig.	DW	Orig.	DW	Orig.	DW	Orig.	DW
Aus	0.702603	0.716240	0.719668	0.722435	0.701951	0.704358	0.705157	0.701496	0.704962	0.701496	0.710810	0.728320
Aut	0.820710	0.836630	0.796200	0.803815	0.786975	0.794327	0.786860	0.802613	0.786860	0.811033	0.811060	0.810884
Bal	0.629429	0.640138	0.729476	0.749348	0.732841	0.761687	0.738482	0.770278	0.738482	0.770278	0.774985	0.736401
Bre	0.567376	0.578319	0.634659	0.646313	0.623179	0.646313	0.600810	0.634729	0.588398	0.634729	0.589701	0.645958
Car	0.943268	0.948443	0.932535	0.927824	0.928719	0.927872	0.920243	0.927877	0.926251	0.926635	0.916533	0.926687
Cle	0.248948	0.258308	0.280058	0.295306	0.259973	0.258413	0.293663	0.289724	0.289809	0.289917	0.283882	0.280330
Der	0.933121	0.940243	0.940228	0.961383	0.936778	0.957920	0.936778	0.954322	0.947408	0.964807	0.936778	0.961383
Eco	0.689089	0.709777	0.725795	0.762571	0.725949	0.762571	0.718035	0.725051	0.725837	0.762571	0.713954	0.742248
For	0.828292	0.846952	0.825349	0.828133	0.825278	0.828133	0.825278	0.828133	0.830630	0.835996	0.833279	0.838708
Ger	0.341661	0.357723	0.466069	0.434980	0.396502	0.409208	0.403696	0.409456	0.451646	0.447000	0.427741	0.393758
Gla	0.669439	0.688949	0.647603	0.673057	0.647766	0.666862	0.636003	0.653978	0.636003	0.653978	0.649689	0.659420
Ilp	0.262097	0.305629	0.216065	0.226721	0.238407	0.242647	0.180429	0.235833	0.181442	0.235833	0.206331	0.241565
Ion	0.824930	0.853441	0.823892	0.841252	0.817971	0.841252	0.831481	0.861053	0.817503	0.841252	0.817893	0.860350
Iri	0.900000	0.910000	0.930000	0.940000	0.940000	0.940000	0.930000	0.940000	0.930000	0.940000	0.920000	0.940000
Lar	0.374862	0.407169	0.474004	0.487102	0.442262	0.473448	0.446673	0.478389	0.449520	0.473877	0.457554	0.484834
Lym	0.588551	0.660981	0.595195	0.672517	0.555655	0.687514	0.555655	0.687514	0.606499	0.672517	0.610839	0.687333
New	0.900172	0.918576	0.892226	0.919735	0.871220	0.898682	0.871220	0.898682	0.881399	0.898682	0.881399	0.898682
Pen	0.938372	0.960599	0.925236	0.960599	0.932311	0.963631	0.920184	0.959588	0.929280	0.963631	0.932313	0.963632
Pim	0.399610	0.403775	0.422095	0.427874	0.435560	0.440594	0.454705	0.464450	0.434233	0.438100	0.454013	0.456037
Son	0.690623	0.717865	0.611630	0.707875	0.602523	0.668764	0.592539	0.707875	0.551699	0.639571	0.563255	0.669332
Use	0.865785	0.882491	0.889629	0.909868	0.893029	0.906525	0.893029	0.906525	0.903475	0.896415	0.906768	0.906525
Ver	0.757992	0.746559	0.725414	0.757473	0.730356	0.752497	0.690668	0.742709	0.679406	0.736703	0.725011	0.757207
Voi	0.388835	0.398440	0.417993	0.438287	0.430643	0.432064	0.430643	0.444012	0.423672	0.444012	0.468207	0.466775
Vow	0.907778	0.941111	0.887778	0.958889	0.874444	0.956667	0.892222	0.963333	0.891111	0.962222	0.893333	0.962222
Wdb	0.883299	0.905815	0.897762	0.916616	0.901612	0.916616	0.897652	0.916616	0.905766	0.931910	0.897762	0.920544
Win	0.956901	0.965672	0.965293	0.965672	0.930510	0.948785	0.974063	0.965672	0.947987	0.965672	0.956347	0.965672
Wis	0.929167	0.938947	0.923144	0.929495	0.923144	0.932786	0.919840	0.929546	0.926417	0.932736	0.926417	0.929471
Wqr	0.444406	0.460806	0.467338	0.473818	0.498379	0.503247	0.447481	0.473275	0.467242	0.470766	0.454649	0.487886
Yea	0.366865	0.394656	0.458876	0.470496	0.463399	0.473868	0.465367	0.472485	0.464427	0.472457	0.464154	0.473160
Zoo	0.920949	0.934727	0.881819	0.895597	0.907791	0.921569	0.881819	0.895597	0.895597	0.895597	0.855993	0.947367
Avg.	0.689171	0.707633	0.703434	0.723502	0.698504	0.720627	0.694689	0.722209	0.697109	0.720346	0.700069	0.725142

Table 7

Results of Wilcoxon signed-rank test for the comparison between the original method and the proposed approaches when homogeneous ensemble is used (κ).

Comparison	R^+	R^-	Hypothesis	p -value
KNORA-E-DW vs. KNORA-E-Orig.	454.00	11.00	Rejected for KNORA-E-DW at 5%	0.000005**
KNORA-U-DW vs. KNORA-U-Orig.	437.00	28.00	Rejected for KNORA-U-DW at 5%	0.000026**
DES-P-DW vs. DES-P-Orig.	458.50	6.50	Rejected for DES-P-DW at 5%	0.000004**
DES-KL-DW vs. DES-KL-Orig.	453.00	12.00	Rejected for DES-KL-DW at 5%	0.000006**
DES-CV-DW vs. DES-CV-Orig.	443.50	21.50	Rejected for DES-CV-DW at 5%	0.000016**
DES-CS-DW vs. DES-CS-Orig.	430.00	35.00	Rejected for DES-CS-DW at 5%	0.000049**

** Close to the p -value means that statistical differences are found with $\alpha = 0.05$ (95% confidence).

Table 8

Results of classification accuracy obtained for using heterogeneous ensemble. The best result in each dataset for each DES method is highlighted in **bold-face**.

Dataset	KNORA-E		KNORA-U		DES-P		DES-KL		DES-CV		DES-CS	
	Orig.	DW	Orig.	DW	Orig.	DW	Orig.	DW	Orig.	DW	Orig.	DW
Aus	83.62	84.78	86.23	85.80	85.36	85.51	86.38	85.94	86.38	85.94	85.94	85.65
Aut	81.19	84.33	81.21	81.15	78.65	81.77	81.19	81.77	83.08	82.44	81.81	81.81
Bal	88.80	88.96	89.28	89.44	89.44	89.28	88.96	89.28	89.60	89.76	90.24	90.40
Bre	70.84	70.79	68.87	68.08	68.00	68.99	69.04	69.04	69.82	68.95	70.73	70.91
Car	97.92	98.26	98.44	97.34	97.74	96.99	94.50	96.99	96.47	97.16	97.28	96.76
Cle	54.84	54.48	58.56	58.90	57.59	57.90	55.85	57.88	56.50	58.53	55.18	58.89
Der	96.92	97.20	97.48	97.48	97.21	97.21	97.48	97.48	97.21	97.21	97.20	97.20
Eco	79.77	82.15	82.46	83.35	81.26	82.75	81.55	81.85	82.74	82.45	82.74	83.94
For	90.06	90.63	90.25	90.06	89.87	90.06	90.25	90.44	90.25	90.44	89.68	89.87
Ger	73.40	74.30	75.50	76.10	75.60	75.40	73.20	75.20	73.50	75.30	74.70	76.20
Gla	73.37	76.63	72.91	74.78	70.11	72.90	69.18	73.85	70.11	74.32	71.50	74.32
Ilp	69.78	69.78	70.81	71.68	72.54	72.19	71.50	71.16	71.85	71.33	71.85	71.85
Ion	92.32	91.75	89.48	89.19	90.33	89.48	89.76	89.19	90.62	90.05	90.90	90.62
Iri	95.33	95.33	95.33	95.33	95.33	95.33	95.33	95.33	95.33	95.33	95.33	95.33
Lar	69.64	71.63	76.76	75.62	76.20	76.48	75.91	75.91	75.06	75.91	74.76	75.91
Lym	81.08	80.37	82.39	82.34	83.79	81.70	83.13	81.03	82.39	83.06	80.41	85.06
New	96.74	97.21	95.35	95.81	95.35	95.81	94.88	95.81	96.28	96.28	95.81	96.28
Pen	98.00	98.27	97.45	97.73	97.27	97.91	97.09	97.45	97.27	97.36	97.18	97.82
Pim	70.31	70.18	76.04	75.39	76.56	76.04	75.91	75.65	75.78	75.52	77.21	75.00
Son	86.98	87.47	84.59	85.55	83.65	83.64	84.09	84.58	86.03	86.03	87.44	86.49
Use	93.02	93.53	93.29	91.54	92.29	91.55	92.79	92.55	93.28	92.04	94.02	92.29
Ver	83.55	82.90	83.55	82.58	82.26	81.94	82.58	81.94	83.87	82.58	81.94	82.26
Voi	74.77	76.04	78.17	78.17	79.87	80.29	79.02	78.59	78.60	78.18	78.16	77.75
Vow	97.88	99.09	96.77	97.68	94.75	97.68	95.05	97.37	94.85	97.37	96.87	98.89
Wdb	96.66	96.66	97.36	97.19	97.19	97.01	96.66	96.84	97.19	97.19	96.66	97.01
Win	97.75	98.32	97.75	98.32	98.30	97.75	97.75	98.32	97.75	97.75	97.75	98.32
Wis	96.78	97.07	96.64	96.93	96.93	96.93	96.93	96.78	96.93	96.78	97.22	97.22
Wqr	63.29	64.79	63.60	65.73	67.79	68.23	62.23	65.29	63.10	65.60	64.60	67.11
Yea	53.37	55.86	61.66	61.39	61.12	61.66	61.73	62.13	60.85	61.80	61.32	61.19
Zoo	95.05	95.05	95.05	96.00	96.00	96.00	95.05	96.00	95.05	96.00	95.05	96.00
Avg.	83.43	84.13	84.44	84.56	84.28	84.55	83.83	84.39	84.26	84.62	84.38	84.94

Table 9

Results of Wilcoxon signed-rank test for the comparison between the original method and the proposed approaches when heterogeneous ensemble is used (classification accuracy).

Comparison	R^+	R^-	Hypothesis	p -value
KNORA-E-DW vs. KNORA-E-Orig.	381.00	84.00	Rejected for KNORA-E-DW at 5%	0.002030**
KNORA-U-DW vs. KNORA-U-Orig.	260.50	204.50	Not Rejected	0.516548
DES-P-DW vs. DES-P-Orig.	269.00	196.00	Not Rejected	0.427880
DES-KL-DW vs. DES-KL-Orig.	329.00	136.00	Rejected for DES-KL-DW at 5%	0.042338**
DES-CV-DW vs. DES-CV-Orig.	278.50	186.50	Not Rejected	0.253098
DES-CS-DW vs. DES-CS-Orig.	331.00	134.00	Rejected for DES-CS-DW at 5%	0.042162**

** Close to the p -value means that statistical differences are found with $\alpha = 0.05$ (95% confidence).

Table 10
Average kappa results obtained for using heterogeneous ensemble. The best result in each dataset for each DES method is highlighted in **bold-face**.

Dataset	KNORA-E		KNORA-U		DES-P		DES-KL		DES-CV		DES-CS	
	Orig.	DW	Orig.	DW	Orig.	DW	Orig.	DW	Orig.	DW	Orig.	DW
Aus	0.670262	0.692546	0.720148	0.711617	0.703061	0.706269	0.722082	0.713863	0.722922	0.714610	0.715186	0.709450
Aut	0.753185	0.793975	0.750620	0.750407	0.715105	0.758008	0.749651	0.758008	0.776180	0.767481	0.760914	0.759743
Bal	0.798622	0.797422	0.801227	0.804159	0.804156	0.801179	0.795274	0.801207	0.807100	0.810071	0.819015	0.821961
Bre	0.647922	0.647451	0.624445	0.614838	0.613603	0.625788	0.625930	0.625930	0.635802	0.625266	0.646683	0.648765
Car	0.954152	0.962314	0.965822	0.942392	0.950102	0.934894	0.873030	0.934346	0.921168	0.938450	0.939907	0.930126
Cle	0.261427	0.266176	0.282861	0.291604	0.248379	0.263422	0.169378	0.255329	0.221713	0.274846	0.181156	0.284497
Der	0.961415	0.964885	0.968419	0.968419	0.964944	0.964944	0.968419	0.968419	0.964944	0.964944	0.964895	0.964884
Eco	0.716875	0.751197	0.751328	0.764938	0.732176	0.755735	0.735834	0.743777	0.756122	0.752304	0.755253	0.773810
For	0.860797	0.868594	0.863168	0.860688	0.857519	0.860534	0.863209	0.866038	0.863340	0.866189	0.855291	0.858143
Ger	0.252963	0.296525	0.288726	0.320882	0.313976	0.315253	0.180670	0.275281	0.203847	0.282153	0.247113	0.326111
Gla	0.635292	0.678894	0.620416	0.647321	0.577633	0.619585	0.562520	0.633187	0.577044	0.639465	0.598877	0.640417
Ilp	0.199536	0.227933	0.022174	0.094580	0.192256	0.199856	0.010271	0.022511	0.027121	0.025915	0.027121	0.113776
Ion	0.826689	0.812601	0.756441	0.750707	0.777359	0.757585	0.763472	0.750092	0.784728	0.770615	0.791447	0.785592
Iri	0.930000	0.930000	0.930000	0.930000	0.930000	0.930000	0.930000	0.930000	0.930000	0.930000	0.930000	0.930000
Lar	0.420260	0.455639	0.529098	0.508016	0.511198	0.521383	0.502687	0.508488	0.490560	0.511271	0.487643	0.513905
Lym	0.626438	0.615804	0.646869	0.644727	0.671592	0.628287	0.659285	0.616769	0.646869	0.662769	0.614163	0.705963
New	0.927091	0.937306	0.892711	0.903512	0.892711	0.903512	0.881270	0.903512	0.915711	0.915711	0.904910	0.915711
Pen	0.977775	0.980805	0.971712	0.974744	0.969691	0.976763	0.967670	0.971712	0.969692	0.970703	0.968682	0.975753
Pim	0.345636	0.327118	0.426621	0.416766	0.450666	0.439199	0.414858	0.410490	0.416223	0.412772	0.459792	0.418782
Son	0.736822	0.746050	0.684831	0.704808	0.665532	0.665560	0.675210	0.685441	0.715018	0.715017	0.745033	0.725635
Use	0.903946	0.910547	0.906656	0.882381	0.892655	0.882309	0.899804	0.896488	0.906648	0.889419	0.917062	0.892853
Ver	0.736624	0.725353	0.733956	0.719096	0.713205	0.709280	0.717441	0.708784	0.739118	0.719537	0.706300	0.713739
Voi	0.409022	0.416247	0.425900	0.425658	0.459365	0.474155	0.440636	0.432537	0.444174	0.424470	0.435734	0.423626
Vow	0.976667	0.990000	0.964444	0.974444	0.942222	0.974444	0.945556	0.971111	0.943333	0.971111	0.965556	0.987778
Wdb	0.928443	0.928443	0.943006	0.939289	0.939141	0.935425	0.927437	0.931381	0.939149	0.939149	0.927673	0.935425
Win	0.965692	0.974530	0.965692	0.974530	0.974065	0.965692	0.965692	0.974530	0.965692	0.965692	0.965692	0.974530
Wis	0.929149	0.935853	0.926243	0.932427	0.932635	0.932438	0.932637	0.929121	0.932615	0.929186	0.939028	0.939016
Wqr	0.418157	0.442077	0.399642	0.438987	0.468219	0.479419	0.369942	0.428898	0.387716	0.435917	0.411118	0.462167
Yea	0.395524	0.427038	0.497597	0.494526	0.487988	0.496482	0.497978	0.504348	0.485914	0.499241	0.492116	0.491743
Zoo	0.934110	0.934466	0.934110	0.946837	0.946837	0.946837	0.933794	0.946837	0.933794	0.946837	0.934110	0.946837
Avg.	0.703350	0.714593	0.706496	0.711110	0.709933	0.714141	0.689388	0.703281	0.700809	0.709037	0.703582	0.719025

Table 11

Results of Wilcoxon signed-rank test for the comparison between the original method and the proposed approaches when heterogeneous ensemble is used (kappa).

Comparison	R^+	R^-	Hypothesis	p-value
KNORA-E-DW vs. KNORA-E-Orig.	378.50	86.50	Rejected for KNORA-E-DW at 5%	0.003074**
KNORA-U-DW vs. KNORA-U-Orig.	272.50	192.50	Not Rejected	0.362371
DES-P-DW vs. DES-P-Orig.	299.00	166.00	Not Rejected	0.178496*
DES-KL-DW vs. DES-KL-Orig.	349.00	116.00	Rejected for DES-KL-DW at 5%	0.016283**
DES-CV-DW vs. DES-CV-Orig.	275.50	189.50	Not Rejected	0.275832
DES-CS-DW vs. DES-CS-Orig.	333.50	131.50	Rejected for DES-CS-DW at 5%	0.035008**

** Close to the p-value means that statistical differences are found with $\alpha = 0.05$ (95% confidence).

are presented in Table 12, in which we can obtain the fact that the behavior of our proposed method is better than the well-known ensemble methods in most of datasets. In addition, we also carry out the Wilcoxon signed-rank test for comparing the proposed method and the representative approaches. The results of the tests are shown in Table 13. From this table, we can conclude that the superiority of the proposed method stands out with respect to the well-known methods.

6. Conclusions

In this paper, we have presented a novel dynamic weighting scheme with the aim to improve the classification ability of DES systems. In order to do so, we have proposed to use the neighborhood of each class of the instance needed to be classified to define the weights to dynamically adjust the decision boundary toward the correct class.

The observed results have shown the usefulness and robustness of the proposed novel methodology in many real-world applications. The influence with respect to the way to generate candidate classifiers in our scenario (i.e. homogeneous ensemble and heterogeneous ensemble) has been analyzed. In addition, we have shown the positive synergy existing between the proposed dynamic weighting scheme and dynamic selection for six state-of-the-art DES approaches. Furthermore, the superiority of the DES with the

Table 12

Average accuracy and kappa test results of representative approaches (Bagging, AdaBoost, RF) and our method (DES-DW). The best result in each dataset for each performance measure is highlighted in **bold-face**.

Dataset	Accuracy				Kappa			
	Bagging	AdaBoost	RF	DES-DW	Bagging	AdaBoost	RF	DES-DW
Aus	85.36	84.93	85.51	86.52	0.705157	0.695817	0.703005	0.728320
Aut	83.71	82.48	86.17	85.58	0.787734	0.769914	0.818257	0.810884
Bal	85.76	84.96	83.36	86.24	0.738482	0.725451	0.691533	0.747985
Bre	69.80	69.00	72.55	70.64	0.635235	0.625650	0.668704	0.645958
Car	96.30	96.53	71.06	96.59	0.920201	0.925201	0.061421	0.926687
Cle	56.80	55.82	56.52	56.48	0.293663	0.278460	0.207351	0.280330
Der	95.81	95.81	96.93	96.92	0.947408	0.947399	0.961346	0.961383
Eco	80.37	80.37	80.67	81.55	0.726825	0.726850	0.722975	0.742248
For	87.57	87.76	89.10	88.52	0.825278	0.828229	0.846790	0.838708
Ger	79.20	78.00	71.50	76.30	0.461416	0.439818	0.082946	0.393758
Gla	74.32	72.92	74.78	75.24	0.647766	0.628391	0.648845	0.659420
Ilp	71.51	70.99	73.58	72.89	0.196558	0.218502	0.214718	0.241565
Ion	92.31	92.03	92.03	93.74	0.831481	0.823469	0.824154	0.860350
Iri	95.33	96.67	94.00	96.00	0.930000	0.950000	0.910000	0.940000
Lar	74.21	71.66	72.78	74.48	0.488664	0.437545	0.456895	0.484834
Lym	77.77	79.10	83.08	83.77	0.568232	0.599364	0.665113	0.687333
New	93.95	93.49	97.21	95.35	0.871220	0.859841	0.938620	0.898682
Pen	93.00	93.45	95.64	96.73	0.922205	0.927262	0.951506	0.963632
Pim	75.52	76.56	74.99	76.04	0.454705	0.468477	0.417288	0.456037
Son	78.86	76.45	81.23	83.67	0.573668	0.522260	0.617763	0.669332
Use	92.27	92.77	92.77	93.27	0.893029	0.899764	0.899584	0.906525
Ver	80.65	82.26	80.32	84.84	0.690668	0.714694	0.680747	0.757207
Voi	76.47	78.17	78.59	79.00	0.422221	0.466200	0.458011	0.466775
Vow	90.00	90.30	95.05	96.57	0.890000	0.893333	0.945556	0.962222
Wdb	95.78	96.13	95.26	96.31	0.908971	0.916683	0.897784	0.920544
Win	95.43	96.02	98.87	97.73	0.930510	0.939596	0.982902	0.965672
Wis	96.78	96.50	97.67	96.78	0.929564	0.922853	0.949012	0.929471
Wqr	67.73	67.10	68.23	68.29	0.476734	0.468626	0.478544	0.487886
Yea	59.37	61.05	55.06	59.64	0.468678	0.491392	0.406509	0.473160
Zoo	91.10	93.10	95.05	96.00	0.881819	0.907958	0.934027	0.947367
Avg.	83.10	83.08	82.99	84.72	0.700603	0.700633	0.668064	0.725142

Table 13

Results of Wilcoxon signed-rank test for the comparison between the original method and the state-of-the-art methods.

Performance measure	Comparison	R^+	R^-	Hypothesis	p -value
Accuracy	DES-DW vs. Bagging	435.50	29.50	Rejected for DES-DW at 5%	0.000038**
	DES-DW vs. AdaBoost	422.00	43.00	Rejected for DES-DW at 5%	0.000097**
	DES-DW vs. RF	363.00	102.00	Rejected for DES-DW at 5%	0.007271**
Kappa	DES-DW vs. Bagging	422.00	43.00	Rejected for DES-DW at 5%	0.000097**
	DES-DW vs. AdaBoost	411.00	54.00	Rejected for DES-DW at 5%	0.000241**
	DES-DW vs. RF	396.00	69.00	Rejected for DES-DW at 5%	0.000771**

** Close to the p -value means that statistical differences are found with $\alpha = 0.05$ (95% confidence).

proposed dynamic combination method also stands out when compared with these well-known ensemble approaches.

The main contributions of this paper with respect to previous works can be summarized as follows.

- We argue in this paper that the information of the local region surrounding the query sample is useful for the combination of selected classifiers when constructing DES classification systems.
- In order to enhance the classification ability of DES systems, we propose a distance-based dynamic weighting scheme for the combination of selected classifiers.
- We carry out extensive experiments on 30 real-world datasets to show the effectiveness and robustness of the proposed method.

We believe that this work draws the future research line for the combination of DES and dynamic weighting framework considering the local information. Furthermore, it would be interesting to explore more effective strategies to compute the weights, such as clustering. In addition, considering the relative worse performance obtained in the scenario of the heterogeneous ensemble, it is worth mentioning that there might be another direction to develop a more appropriate method to estimate the confidence for the heterogeneous ensemble.

Acknowledgments

The authors would like to thank the (anonymous) reviewers for their constructive comments. This work is financially supported by the National Science Foundation of China (NSFC Proj. 71801065, 71831006 and 71771070).

References

- Almeida, P. R., Oliveira, L. S., Britto, A. S., & Sabourin, R. (2018). Adapting dynamic classifier selection for concept drift. *Expert Systems with Applications*, 104, 67–85.
- Amozegar, M., & Khorasani, K. (2016). An ensemble of dynamic neural network identifiers for fault detection and isolation of gas turbine engines. *Neural Networks*, 76, 106–121.
- Britto, A. S., Jr, Sabourin, R., & Oliveira, L. E. S. (2014). Dynamic selection of classifiers - A comprehensive review. *Pattern Recognition*, 47(11), 3665–3680.
- Bashbaghi, S., Granger, E., Sabourin, R., & Bilodeau, G. A. (2017). Dynamic ensembles of exemplar-SVMs for still-to-video face recognition. *Pattern Recognition*, 69, 61–81.
- Berger, J. O. (2013). *Statistical decision theory and Bayesian analysis*. Springer Science & Business Media.
- Breiman, L. (1996). Bagging predictors. *Machine Learning*, 24(2), 123–140.
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.
- Brun, A. L., Britto, A. S., Oliveira, L. S., Enembreck, F., & Sabourin, R. (2016). Contribution of data complexity features on dynamic classifier selection. *International joint conference on neural networks (IJCNN)* 4396–4403.
- Capozzoli, A., Lauro, F., & Khan, I. (2015). Fault detection analysis using data mining techniques for a cluster of smart office buildings. *Expert Systems with Applications*, 42(9), 4324–4338.
- Cavalin, P. R., Sabourin, R., & Suen, C. Y. (2012). LoGID: An adaptive framework combining local and global incremental learning for dynamic selection of ensembles of HMMs. *Pattern Recognition*, 45(9), 3544–3556.
- Cevikalp, H., & Polikar, R. (2008). Local classifier weighting by quadratic programming. *IEEE Transactions on Neural Networks*, 19(10), 1832–1838.
- Cohen, J. (1960). A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1), 37–46.
- Cruz, R. M., Oliveira, D. V., Cavalcanti, G. D., & Sabourin, R. (2019). FIRE-DES++: Enhanced online pruning of base classifiers for dynamic ensemble selection. *Pattern Recognition*, 85, 149–160.
- Cruz, R. M., Sabourin, R., & Cavalcanti, G. D. (2017). META-DES. Oracle: Meta-learning and feature selection for dynamic ensemble selection. *Information Fusion*, 38, 84–103.
- Cruz, R. M., Sabourin, R., & Cavalcanti, G. D. (Sabourin, Cavalcanti, 2018a). Dynamic classifier selection: Recent advances and perspectives. *Information Fusion*, 41, 195–216.
- Cruz, R. M., Sabourin, R., Cavalcanti, G. D., & Ren, T. I. (2015). META-DES: A dynamic ensemble selection framework using meta-learning. *Pattern Recognition*, 48(5), 1925–1935.
- Cruz, R. M. O., Sabourin, R., & Cavalcanti, G. D. C. (Sabourin, Cavalcanti, 2018b). Prototype selection for dynamic classifier and ensemble selection. *Neural Computing and Applications*, 29(2), 447–457.
- Derrac, J., García, S., Molina, D., & Herrera, F. (2011). A practical tutorial on the use of nonparametric statistical tests as a methodology for comparing evolutionary and swarm intelligence algorithms. *Swarm and Evolutionary Computation*, 1(1), 3–18.
- Didaci, L., Giacinto, G., Roli, F., & Marcialis, G. L. (2005). A study on the performances of dynamic classifier selection based on local accuracy estimation. *Pattern Recognition*, 38(11), 2188–2191.
- Dietterich, T. G. (2000). *Ensemble methods in machine learning*. Berlin Heidelberg: Springer.
- Duin, R. P. W. (2002). The combining classifier: To train or not to train? *Proceedings of the 16th international conference on pattern recognition* 765–770.
- Duin, R. P. W., Juszczak, P., Paclik, P., Pekalska, E., de Ridder, D., Tax, D. M. J., & Verzakov, S. (2007). PR-Tools 4.1: A matlab toolbox for pattern recognition. <http://prtools.org>.

- Feng, X., Xiao, Z., Zhong, B., Dong, Y., & Qiu, J. (2019). Dynamic weighted ensemble classification for credit scoring using markov chain. *Applied Intelligence*, 49, 555–568.
- Freund, Y., & Schapire, R. E. (1995). A decision-theoretic generalization of on-line learning and an application to boosting. *European conference on computational learning theory*. Springer 23–37.
- Freund, Y., Schapire, R. E., et al. (1996). Experiments with a new boosting algorithm. *ICMLvol. 96. ICML* 148–156.
- Galar, M., Fernández, A., Barrenechea, E., & Herrera, F. (2015). DRCW-OVO: Distance-based relative competence weighting combination for one-vs-one strategy in multi-class problems. *Pattern Recognition*, 48(1), 28–42.
- García, S., Fernández, A., Luengo, J., & Herrera, F. (2010). Advanced nonparametric tests for multiple comparisons in the design of experiments in computational intelligence and data mining: Experimental analysis of power. *Information Sciences*, 180(10), 2044–2064.
- García, S., Zhang, Z. L., Altalhi, A., Alshomrani, S., & Herrera, F. (2018). Dynamic ensemble selection for multi-class imbalanced datasets. *Information Sciences*, 445–446, 22–37.
- Giacinto, G., & Roli, F. (1999). Methods for dynamic classifier selection. *International conference on image analysis and processing* 659–664.
- Giacinto, G., & Roli, F. (2000). Dynamic classifier selection. *Multiple Classifier Systems*, 177–189.
- Giacinto, G., & Roli, F. (2001). Dynamic classifier selection based on multiple classifier behaviour. *Pattern Recognition*, 34, 1879–1881.
- Goswami, G., Mittal, P., Majumdar, A., Vatsa, M., & Singh, R. (2016). Group sparse representation based classification for multi-feature multimodal biometrics. *Information Fusion*, 32, 3–12.
- Štefka, D., & Holeňa, M. (2015). Dynamic classifier aggregation using interaction-sensitive fuzzy measures. *Fuzzy Sets and Systems*, 270, 25–52.
- Hansen, L. K., Lüsberg, C., & Salamon, P. (1997). The error-reject tradeoff. *Open Systems & Information Dynamics*, 4(2), 159–184.
- Ho, T. K. (1998). The random subspace method for constructing decision forests. *IEEE transactions on pattern analysis and machine intelligence* 20. *IEEE transactions on pattern analysis and machine intelligence* 832–844.
- Ho, T. K., Hull, J. J., & Srihari, S. N. (1994). Decision combination in multiple classifier systems. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 16(1), 66–75.
- Huang, Y. S., & Suen, C. Y. (1995). A method of combining multiple experts for the recognition of unconstrained handwritten numerals. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 17(1), 90–94.
- Jacobs, R. A., Jordan, M. I., Nowlan, S. J., & Hinton, G. E. (1991). Adaptive mixtures of local experts. *Neural Computation*, 3, 79–87.
- Juefei-Xu, F., & Savvides, M. (2016). Multi-class fukunaga koontz discriminant analysis for enhanced face recognition. *Pattern Recognition*, 52, 186–205.
- Kessentini, Y., Burger, T., & Paquet, T. (2015). A Dempster-Shafer theory based combination of handwriting recognition systems with multiple rejection strategies. *Pattern Recognition*, 48(2), 534–544.
- Kittler, J., Hatef, M., Duin, R. P. W., & Matas, J. (1998). On combining classifiers. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 20(3), 226–239.
- Ko, A. H., Sabourin, R., & Britto, A. S., Jr (2008). From dynamic classifier selection to dynamic ensemble selection. *Pattern Recognition*, 41(5), 1718–1731.
- Krawczyk, B., Galar, M., Woźniak, M., Bustince, H., & Herrera, F. (2018). Dynamic ensemble selection for multi-class classification with one-class classifiers. *Pattern Recognition*, 83, 34–51.
- Kuncheva, L. I., Bezdek, J. C., & Duin, R. P. (2001). Decision templates for multiple classifier fusion: An experimental comparison. *Pattern Recognition*, 34(2), 299–314.
- Lin, C., Chen, W., Qiu, C., Wu, Y., Krishnan, S., & Zou, Q. (2014). LibD3C: Ensemble classifiers with a clustering and dynamic selection strategy. *Neurocomputing*, 123, 424–435.
- Lu, Y. (1996). Knowledge integration in a multiple classifier system. *Applied Intelligence*, 6(2), 75–86.
- Markatopoulou, F., Tsoumakas, G., & Vlahavas, I. (2015). Dynamic ensemble pruning based on multi-label classification. *Neurocomputing*, 150, 501–512.
- Moreno-Torres, J. G., Saez, J. A., & Herrera, F. (2012). Study on the impact of partition-induced dataset shift on k-fold cross-validation. *IEEE Transactions on Neural Networks & Learning Systems*, 23(8), 1304–1312.
- Oliveira, D. V., Cavalcanti, G. D., & Sabourin, R. (2017). Online pruning of base classifiers for dynamic ensemble selection. *Pattern Recognition*, 72, 44–58.
- Onan, A., Korukoğlu, S., & Bulut, H. (2017). A hybrid ensemble pruning approach based on consensus clustering and multi-objective evolutionary algorithm for sentiment classification. *Information Processing and Management*, 53, 814–833.
- Peker, M. (2016). A decision support system to improve medical diagnosis using a combination of k-medoids clustering based attribute weighting and svm. *Journal of Medical Systems*, 40(5), 116–131.
- Rahman, A., & Verma, B. (2013). Effect of ensemble classifier composition on offline cursive character recognition. *Information Processing and Management*, 49(4), 852–864.
- Rehman, A., Javed, K., & Babri, H. A. (2017). Feature selection based on a normalized difference measure for text classification. *Information Processing and Management*, 53, 473–489.
- Roy, A., Cruz, R. M., Sabourin, R., & Cavalcanti, G. D. (2018). A study on combining dynamic selection and data preprocessing for imbalance learning. *Neurocomputing*, 286, 179–192.
- Santos, E. M. D., Sabourin, R., & Maupin, P. (2007). Ambiguity-guided dynamic selection of ensemble of classifiers. *International conference on information fusion* 1–8.
- Santos, E. M. D., Sabourin, R., & Maupin, P. (2008). A dynamic overproduce-and-choose strategy for the selection of classifier ensembles. *Pattern Recognition*, 41(10), 2993–3009.
- Santos, E. M. D., Sabourin, R., & Maupin, P. (2009). Overfitting cautious selection of classifier ensembles with genetic algorithms. *Information Fusion*, 10(2), 150–162.
- Stanfill, C., & Waltz, D. (1986). Toward memory-based reasoning. *Communications of the ACM*, 29(12), 1213–1228.
- Sun, S. (2010). Local within-class accuracies for weighting individual outputs in multiple classifier systems. *Pattern Recognition Letters*, 31(2), 119–124.
- Tsybal, A., Pechenizkiy, M., Cunningham, P., & Puuronen, S. (2008). Dynamic integration of classifiers for handling concept drift. *Information Fusion*, 9(1), 56–68.
- Vriesmann, L. M., Britto, A. S., Oliveira, L. S., Koerich, A. L., & Sabourin, R. (2015). Combining overall and local class accuracies in an oracle-based method for dynamic ensemble selection. *International joint conference on neural networks (IJCNN)* 1–7.
- Wilcoxon, F. (1945). Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6), 80–83.
- Wilson, D. R., & Martinez, T. R. (1996). Improved heterogeneous distance functions. *Journal of Artificial Intelligence Research*, 6(1), 1–34.
- Woloszynski, T., & Kurzynski, M. (2011). A probabilistic model of classifier competence for dynamic ensemble selection. *Pattern Recognition*, 44(10–11), 2656–2668.
- Woloszynski, T., Kurzynski, M., Podsiadło, P., & Stachowiak, G. W. (2012). A measure of competence based on random classification for dynamic ensemble selection. *Information Fusion*, 13(3), 207–213.
- Woods, K., Kegelmeyer, W. P., & Bowyer, K. (1997). Combination of multiple classifiers using local accuracy estimates. *IEEE transactions on pattern analysis and machine intelligence* 19. *IEEE transactions on pattern analysis and machine intelligence* 405–410.
- Woźniak, M., Graña, M., & Corchado, E. (2014). A survey of multiple classifier systems as hybrid systems. *Information Fusion*, 16, 3–17.
- Xiao, J., Xie, L., He, C., & Jiang, X. (2012). Dynamic classifier ensemble model for customer classification with imbalanced class distribution. *Expert Systems with Applications*, 39(3), 3668–3675.
- Ye, R., & Dai, Q. (2018). A novel greedy randomized dynamic ensemble selection algorithm. *Neural Processing Letters*, 47(2), 565–599.
- Zhang, Z. L., Luo, X. G., García, S., Tang, J. F., & Herrera, F. (2017). Exploring the effectiveness of dynamic ensemble selection in the one-versus-one scheme. *Knowledge-Based Systems*, 125, 53–63.
- Zhang, Z. L., Luo, X. G., Yu, Y., Yuan, B. W., & Tang, J. F. (2018). Integration of an improved dynamic ensemble selection approach to enhance one-vs-one scheme. *Engineering Applications of Artificial Intelligence*, 74, 43–53.